Construction and universal exactness of the Gabriel–Serre quotient

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Allowed standard prerequisites (editorial).

- Abelian categories and basic homological algebra. Original localization and Serre-subcategory definitions and supporting proofs are included.. The quotient category is constructed, not supplied as a free premise.

1 Localization

In this section we note how Gabriel-Zisman localization interacts with the additive structure on a category.

Lemma 1.1. Let $\mathcal{C}$ be a preadditive category. Let S be a left or right multiplicative system. There exists a unique preadditive structure on $S^{-1}\mathcal{C}$ such that the localization functor $Q : \mathcal{C} \to S^{-1}\mathcal{C}$ is additive.

Proof. We will prove this in the case S is a left multiplicative system. The case where S is a right multiplicative system is dual. Suppose that X, Y are objects of $\mathcal{C}$ and that $\alpha, \beta : X \to Y$ are morphisms in $S^{-1}\mathcal{C}$. According to Categories, Lemma [original label: `categories-lemma-morphisms-left-localization`] we may represent these by pairs $s^{-1}f, s^{-1}g$ with common denominator s. Suppose that we have a pre-additive structure on $S^{-1}\mathcal{C}$ and that Q is additive. Then

$$\begin{aligned} s^{-1}f + s^{-1}g &= Q(s)^{-1} \circ Q(f) + Q(s)^{-1} \circ Q(g) \\ &= Q(s)^{-1} \circ (Q(f) + Q(g)) \\ &= Q(s)^{-1} \circ Q(f + g) \\ &= s^{-1}(f + g). \end{aligned}$$

Hence, the pre-additive structure on $S^{-1}\mathcal{C}$ is unique if it exists. Correspondingly, we define $\alpha + \beta$ to be the equivalence class of $s^{-1}(f+g)$ and in the rest of the proof we show that this is well defined and that composition is bilinear. Once this is done it is clear that Q is an additive functor.

Let us show construction above is well defined. An abstract way of saying this is that filtered colimits of abelian groups agree with filtered colimits of sets and to use Categories, Equation ([original label: `categories-equation-left-localization-morphisms-colimit`]). We can work this out in a bit more detail as follows. Say $s : Y \to Y_1$ and $f, g : X \to Y_1$. Suppose we have a second representation of α, β as $(s')^{-1}f', (s')^{-1}g'$ with $s' : Y \to Y_2$ and $f', g' : X \to Y_2$. By Categories, Remark [original label: `categories-remark-left-localization-morphisms-colimit`] we can find a morphism $s_3 : Y \to Y_3$ and morphisms $a_1 : Y_1 \to Y_3$, $a_2 : Y_2 \to Y_3$ such that $a_1 \circ s = s_3 = a_2 \circ s'$ and also $a_1 \circ f = a_2 \circ f'$ and $a_1 \circ g = a_2 \circ g'$. Hence we see that $s^{-1}(f + g)$ is equivalent to

$$\begin{aligned} s_3^{-1}(a_1 \circ (f + g)) &= s_3^{-1}(a_1 \circ f + a_1 \circ g) \\ &= s_3^{-1}(a_2 \circ f' + a_2 \circ g') \\ &= s_3^{-1}(a_2 \circ (f' + g')) \end{aligned}$$

which is equivalent to $(s')^{-1}(f' + g')$.

Fix $s : Y \to Y'$ and $f, g : X \to Y'$ with $\alpha = s^{-1}f$ and $\beta = s^{-1}g$ as morphisms $X \to Y$ in $S^{-1}\mathcal{C}$. To show that composition is bilinear first consider the case of a morphism $\gamma : Y \to Z$ in $S^{-1}\mathcal{C}$. Say $\gamma = t^{-1}h$ for some

$h : Y \to Z'$ and $t : Z \to Z'$ in S. Using LMS2 we choose morphisms $a : Y' \to Z''$ and $t' : Z' \to Z''$ in S such that $a \circ s = t' \circ h$. Picture

$$\begin{array}{ccccc} & & & & Z \\ & & & & \downarrow{\scriptstyle t} \\ & & Y & \xrightarrow{h} & Z' \\ & & \downarrow{\scriptstyle s} & & \downarrow{\scriptstyle t'} \\ X & \xrightarrow{f,g} & Y' & \xrightarrow{a} & Z'' \end{array}$$

Then $\gamma \circ \alpha = (t' \circ t)^{-1}(a \circ f)$ and $\gamma \circ \beta = (t' \circ t)^{-1}(a \circ g)$. Hence we see that $\gamma \circ (\alpha + \beta)$ is represented by $(t' \circ t)^{-1}(a \circ (f + g)) = (t' \circ t)^{-1}(a \circ f + a \circ g)$ which represents $\gamma \circ \alpha + \gamma \circ \beta$.

Finally, assume that $\delta : W \to X$ is another morphism of $S^{-1}\mathcal{C}$. Say $\delta = r^{-1}i$ for some $i : W \to X'$ and $r : X \to X'$ in S. We claim that we can find a morphism $s' : Y' \to Y''$ in S and morphisms $a'', b'' : X' \to Y''$ such that the following diagram commutes

$$\begin{array}{ccccc} & & & & Y \\ & & & & \downarrow{\scriptstyle s} \\ & & X & \xrightarrow{f,g,f+g} & Y' \\ & & \downarrow{\scriptstyle r} & & \downarrow{\scriptstyle s'} \\ W & \xrightarrow{i} & X' & \xrightarrow{a'',b'',a''+b''} & Y'' \end{array}$$

Namely, using LMS2 we can first choose $s_1 : Y' \to Y_1$, $s_2 : Y' \to Y_2$ in S and $a : X' \to Y_1$, $b : X' \to Y_2$ such that $a \circ r = s_1 \circ f$ and $b \circ r = s_2 \circ f$. Then using that the category Y'/S is filtered (see Categories, Remark [original label: `categories-remark-left-localization-morphisms-colimit`]), we can find a $s' : Y' \to Y''$ and morphisms $a' : Y_1 \to Y''$, $b' : Y_2 \to Y''$ such that $s' = a' \circ s_1$ and $s' = b' \circ s_2$. Setting $a'' = a' \circ a$ and $b'' = b' \circ b$ works. At this point we see that the compositions $\alpha \circ \delta$ and $\beta \circ \delta$ are represented by $(s' \circ s)^{-1}(a'' \circ i)$ and $(s' \circ s)^{-1}(b'' \circ i)$. Hence $\alpha \circ \delta + \beta \circ \delta$ is represented by $(s' \circ s)^{-1}(a'' \circ i + b'' \circ i) = (s' \circ s)^{-1}((a'' + b'') \circ i)$ which by the diagram again is a representative of $(\alpha + \beta) \circ \delta$. □

Lemma 1.2. Let $\mathcal{C}$ be an additive category. Let S be a left or right multiplicative system. Then $S^{-1}\mathcal{C}$ is an additive category and the localization functor $Q : \mathcal{C} \to S^{-1}\mathcal{C}$ is additive.

Proof. By Lemma 1.1 we see that $S^{-1}\mathcal{C}$ is preadditive and that Q is additive. By Lemma [original label: `lemma-additive-additive`] we conclude that $S^{-1}\mathcal{C}$ has a zero object and direct sums. □

The following lemma describes the "kernel" of the localization functor in case we invert a multiplicative system.

Lemma 1.3. Let $\mathcal{C}$ be an additive category. Let S be a multiplicative system. Let X be an object of $\mathcal{C}$. The following are equivalent

1. $Q(X) = 0$ in $S^{-1}\mathcal{C}$,
2. there exists $Y \in \mathrm{Ob}(\mathcal{C})$ such that $0 : X \to Y$ is an element of S, and
3. there exists $Z \in \mathrm{Ob}(\mathcal{C})$ such that $0 : Z \to X$ is an element of S.

Proof. If (2) holds we see that $0 = Q(0) : Q(X) \to Q(Y)$ is an isomorphism. In the additive category $S^{-1}\mathcal{C}$ this implies that $Q(X) = 0$. Hence (2) $\Rightarrow$ (1). Similarly, (3) $\Rightarrow$ (1). Suppose that $Q(X) = 0$. This implies that the morphism $f : 0 \to X$ is transformed into an isomorphism in $S^{-1}\mathcal{C}$. Hence by Categories, Lemma [original label: `categories-lemma-what-gets-inverted`] there exists a morphism $g : Z \to 0$ such that $fg \in S$. This proves (1) $\Rightarrow$ (3). Similarly, (1) $\Rightarrow$ (2). □

Lemma 1.4. Let $\mathcal{A}$ be an abelian category.

1. If S is a left multiplicative system, then the category $S^{-1}\mathcal{A}$ has cokernels and the functor $Q : \mathcal{A} \to S^{-1}\mathcal{A}$ commutes with them.
2. If S is a right multiplicative system, then the category $S^{-1}\mathcal{A}$ has kernels and the functor $Q : \mathcal{A} \to S^{-1}\mathcal{A}$ commutes with them.
3. If S is a multiplicative system, then the category $S^{-1}\mathcal{A}$ is abelian and the functor $Q : \mathcal{A} \to S^{-1}\mathcal{A}$ is exact.

Proof. Assume S is a left multiplicative system. Let $a : X \to Y$ be a morphism of $S^{-1}\mathcal{A}$. Then $a = s^{-1}f$ for some $s : Y \to Y'$ in S and $f : X \to Y'$. Since $Q(s)$ is an isomorphism we see that the existence of $\mathrm{Coker}(a : X \to Y)$ is equivalent to the existence of $\mathrm{Coker}(Q(f) : X \to Y')$. Since $\mathrm{Coker}(Q(f))$ is the coequalizer of 0 and $Q(f)$ we see that $\mathrm{Coker}(Q(f))$ is represented by $Q(\mathrm{Coker}(f))$ by Categories, Lemma [original label: `categories-lemma-left-localization-limits`]. This proves (1).

Part (2) is dual to part (1).

If S is a multiplicative system, then S is both a left and a right multiplicative system. Thus we see that $S^{-1}\mathcal{A}$ has kernels and cokernels and Q commutes with kernels and cokernels. To finish the proof of (3) we have to show that Coim = Im in $S^{-1}\mathcal{A}$. Again using that any arrow in $S^{-1}\mathcal{A}$ is isomorphic to an arrow $Q(f)$ we see that the result follows from the result for $\mathcal{A}$. □

2 Jordan-Hölder

The Jordan-Hölder lemma is Lemma 2.7. First we state some definitions.

Definition 2.1. Let $\mathcal{A}$ be an abelian category. An object A of $\mathcal{A}$ is said to be simple if it is nonzero and the only subobjects of A are 0 and A.

Definition 2.2. Let $\mathcal{A}$ be an abelian category.

1. We say an object A of $\mathcal{A}$ is Artinian if and only if it satisfies the descending chain condition for subobjects.
2. We say $\mathcal{A}$ is Artinian if every object of $\mathcal{A}$ is Artinian.

Definition 2.3. Let $\mathcal{A}$ be an abelian category.

1. We say an object A of $\mathcal{A}$ is Noetherian if and only if it satisfies the ascending chain condition for subobjects.
2. We say $\mathcal{A}$ is Noetherian if every object of $\mathcal{A}$ is Noetherian.

Lemma 2.4. Let $\mathcal{A}$ be an abelian category. Let $0 \to A_1 \to A_2 \to A_3 \to 0$ be a short exact sequence of $\mathcal{A}$. Then A_2 is Artinian if and only if A_1 and A_3 are Artinian.

Proof. Omitted. □

Lemma 2.5. Let $\mathcal{A}$ be an abelian category. Let $0 \to A_1 \to A_2 \to A_3 \to 0$ be a short exact sequence of $\mathcal{A}$. Then A_2 is Noetherian if and only if A_1 and A_3 are Noetherian.

Proof. Omitted. □

Lemma 2.6. Let $\mathcal{A}$ be an abelian category. Let A be an object of $\mathcal{A}$. The following are equivalent

1. A is Artinian and Noetherian, and
2. there exists a filtration $0 \subset A_1 \subset A_2 \subset \ldots \subset A_n = A$ by subobjects such that A_i/A_{i-1} is simple for $i = 1, \ldots, n$.

Proof. Assume (1). If A is zero, then (2) holds. If A is not zero, then there exists a smallest nonzero object $A_1 \subset A$ by the Artinian property. Of course A_1 is simple. If $A_1 = A$, then we are done. If not, then we can find $A_1 \subset A_2 \subset A$ minimal with $A_2 \neq A_1$. Then A_2/A_1 is simple. Continuing in this way, we can find a sequence $0 \subset A_1 \subset A_2 \subset \ldots$ of subobjects of A such that A_i/A_{i-1} is simple. Since A is Noetherian, we conclude that the process stops. Hence (2) follows.

Assume (2). We will prove (1) by induction on n. If $n = 1$, then A is simple and clearly Noetherian and Artinian. If the result holds for $n - 1$, then we use the short exact sequence $0 \to A_{n-1} \to A_n \to A_n/A_{n-1} \to 0$ and Lemmas 2.4 and 2.5 to conclude for n. □

Lemma 2.7 (Jordan-Hölder). Let $\mathcal{A}$ be an abelian category. Let A be an object of $\mathcal{A}$ satisfying the equivalent conditions of Lemma 2.6. Given two filtrations

$$0 \subset A_1 \subset A_2 \subset \ldots \subset A_n = A \quad \text{and} \quad 0 \subset B_1 \subset B_2 \subset \ldots \subset B_m = A$$

with $S_i = A_i/A_{i-1}$ and $T_j = B_j/B_{j-1}$ simple objects we have $n = m$ and there exists a permutation σ of $\{1, \ldots, n\}$ such that $S_i \cong T_{\sigma(i)}$ for all $i \in \{1, \ldots, n\}$.

Proof. Let j be the smallest index such that $A_1 \subset B_j$. Then the map $S_1 = A_1 \to B_j/B_{j-1} = T_j$ is an isomorphism. Moreover, the object $A/A_1 = A_n/A_1 = B_m/A_1$ has the two filtrations

$$0 \subset A_2/A_1 \subset A_3/A_1 \subset \ldots \subset A_n/A_1$$

and

$$0 \subset (B_1 + A_1)/A_1 \subset \ldots \subset (B_{j-1} + A_1)/A_1 = B_j/A_1 \subset B_{j+1}/A_1 \subset \ldots \subset B_m/A_1$$

We conclude by induction. □

3 Serre subcategories

In [1, Chapter I, Section 1] a notion of a "class" of abelian groups is defined. This notion has been extended to abelian categories by many authors (in slightly different ways). We will use the following variant which is virtually identical to Serre's original definition.

Definition 3.1. [1, Condition (I) on page 259] Let $\mathcal{A}$ be an abelian category.

1. A Serre subcategory of $\mathcal{A}$ is a nonempty full subcategory $\mathcal{C}$ of $\mathcal{A}$ such that given an exact sequence[1]

$$A \to B \to C$$

with $A, C \in \mathrm{Ob}(\mathcal{C})$, then also $B \in \mathrm{Ob}(\mathcal{C})$.

2. A weak Serre subcategory of $\mathcal{A}$ is a nonempty full subcategory $\mathcal{C}$ of $\mathcal{A}$ such that given an exact sequence

$$A_0 \to A_1 \to A_2 \to A_3 \to A_4$$

with A_0, A_1, A_3, A_4 in $\mathcal{C}$, then also A_2 in $\mathcal{C}$.

In some references the second notion is called a "thick" subcategory and in other references the first notion is called a "thick" subcategory. However, it seems that the notion of a Serre subcategory is universally accepted to be the one defined above. Note that in both cases the category $\mathcal{C}$ is abelian and that the inclusion functor $\mathcal{C} \to \mathcal{A}$ is a fully faithful exact functor. Let's characterize these types of subcategories in more detail.

Lemma 3.2. Let $\mathcal{A}$ be an abelian category. Let $\mathcal{C}$ be a subcategory of $\mathcal{A}$. Then $\mathcal{C}$ is a Serre subcategory if and only if the following conditions are satisfied:

1. $0 \in \mathrm{Ob}(\mathcal{C})$,
2. $\mathcal{C}$ is a strictly full subcategory of $\mathcal{A}$,
3. any subobject or quotient of an object of $\mathcal{C}$ is an object of $\mathcal{C}$,
4. if $A \in \mathrm{Ob}(\mathcal{A})$ is an extension of objects of $\mathcal{C}$ then also $A \in \mathrm{Ob}(\mathcal{C})$.

Moreover, a Serre subcategory is an abelian category and the inclusion functor is exact.

Proof. Omitted. □

Lemma 3.3. Let $\mathcal{A}$ be an abelian category. Let $\mathcal{C}$ be a subcategory of $\mathcal{A}$. Then $\mathcal{C}$ is a weak Serre subcategory if and only if the following conditions are satisfied:

1. $0 \in \mathrm{Ob}(\mathcal{C})$,
2. $\mathcal{C}$ is a strictly full subcategory of $\mathcal{A}$,
3. kernels and cokernels in $\mathcal{A}$ of morphisms between objects of $\mathcal{C}$ are in $\mathcal{C}$,
4. if $A \in \mathrm{Ob}(\mathcal{A})$ is an extension of objects of $\mathcal{C}$ then also $A \in \mathrm{Ob}(\mathcal{C})$.

Moreover, a weak Serre subcategory is an abelian category and the inclusion functor is exact.

Proof. Omitted. □

Lemma 3.4. Let $\mathcal{A}$, $\mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be an exact functor. Then the full subcategory of objects C of $\mathcal{A}$ such that $F(C) = 0$ forms a Serre subcategory of $\mathcal{A}$.

[1] By Definition [original label: `definition-exact`] this means $\mathrm{Im}(A \to B) = \mathrm{Ker}(B \to C)$.

Proof. Omitted. □

Definition 3.5. Let $\mathcal{A}$, $\mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be an exact functor. Then the full subcategory of objects C of $\mathcal{A}$ such that $F(C) = 0$ is called the kernel of the functor F, and is sometimes denoted $\mathrm{Ker}(F)$.

Any Serre subcategory of an abelian category is the kernel of an exact functor. In Examples, Section [original label: `examples-section-serre-quotient-modulo-torsion-modules`] we discuss this for Serre's original example of torsion groups.

Lemma 3.6. Let $\mathcal{A}$ be an abelian category. Let $\mathcal{C} \subset \mathcal{A}$ be a Serre subcategory. There exists an abelian category $\mathcal{A}/\mathcal{C}$ and an exact functor

$$F : \mathcal{A} \longrightarrow \mathcal{A}/\mathcal{C}$$

which is essentially surjective and whose kernel is $\mathcal{C}$. The category $\mathcal{A}/\mathcal{C}$ and the functor F are characterized by the following universal property: For any exact functor $G : \mathcal{A} \to \mathcal{B}$ such that $\mathcal{C} \subset \mathrm{Ker}(G)$ there exists a factorization $G = H \circ F$ for a unique exact functor $H : \mathcal{A}/\mathcal{C} \to \mathcal{B}$.

Proof. Consider the set of arrows of $\mathcal{A}$ defined by the following formula

$$S = \{f \in \mathrm{Arrows}(\mathcal{A}) \mid \mathrm{Ker}(f), \mathrm{Coker}(f) \in \mathrm{Ob}(\mathcal{C})\}.$$

We claim that S is a multiplicative system. To prove this we have to check MS1, MS2, MS3, see Categories, Definition [original label: `categories-definition-multiplicative-system`].

It is clear that identities are elements of S. Suppose that $f : A \to B$ and $g : B \to C$ are elements of S. There are exact sequences

$$0 \to \mathrm{Ker}(f) \to \mathrm{Ker}(gf) \to \mathrm{Ker}(g)$$
$$\mathrm{Coker}(f) \to \mathrm{Coker}(gf) \to \mathrm{Coker}(g) \to 0$$

Hence it follows that $gf \in S$. This proves MS1. (In fact, a similar argument will show that S is a saturated multiplicative system, see Categories, Definition [original label: `categories-definition-saturated-multiplicative-system`].)

Consider a solid diagram

$$\begin{array}{ccc} A & \xrightarrow{g} & B \\ {\scriptstyle t}\downarrow & & \vdots\,{\scriptstyle s} \\ C & \overset{f}{\dashrightarrow} & C \amalg_A B \end{array}$$

with $t \in S$. Set $W = C \amalg_A B = \mathrm{Coker}((t, -g) : A \to C \oplus B)$. Then $\mathrm{Ker}(t) \to \mathrm{Ker}(s)$ is surjective and $\mathrm{Coker}(t) \to \mathrm{Coker}(s)$ is an isomorphism. Hence s is an element of S. This proves LMS2 and the proof of RMS2 is dual.

Finally, consider morphisms $f, g : B \to C$ and a morphism $s : A \to B$ in S such that $f \circ s = g \circ s$. This means that $(f - g) \circ s = 0$. In turn this means that $I = \mathrm{Im}(f - g) \subset C$ is a quotient of $\mathrm{Coker}(s)$ hence an object of $\mathcal{C}$. Thus $t : C \to C' = C/I$ is an element of S such that $t \circ (f - g) = 0$, i.e., such that $t \circ f = t \circ g$. This proves LMS3 and the proof of RMS3 is dual.

Having proved that S is a multiplicative system we set $\mathcal{A}/\mathcal{C} = S^{-1}\mathcal{A}$, and we set F equal to the localization functor Q. By Lemma 1.4 the category $\mathcal{A}/\mathcal{C}$ is abelian and F is exact. If X is in the kernel of $F = Q$, then by Lemma 1.3 we see that $0 : X \to Z$ is an element of S and hence X is an object of $\mathcal{C}$, i.e., the kernel of F is $\mathcal{C}$. Finally, if G is as in the statement of the lemma, then G turns every element of S into an isomorphism. Hence we obtain the functor $H : \mathcal{A}/\mathcal{C} \to \mathcal{B}$ from the universal property of localization, see Categories, Lemma [original label: `categories-lemma-properties-left-localization`]. We still have to show the functor H is exact. To do this it suffices to show that H commutes with taking kernels and cokernels, see Lemma [original label: `lemma-exact-functor`]. Let $A \to B$ be a morphism in $\mathcal{A}/\mathcal{C}$. We may represent $A \to B$ as fs^{-1} where $s : A' \to A$ is in S and $f : A' \to B$ an arbitrary morphism of $\mathcal{A}$. Since $F = Q$ maps s to an isomorphism in the quotient category $\mathcal{A}/\mathcal{C}$, it suffices to show that H commutes with taking kernels and cokernels of morphisms $f : A \to B$ of $\mathcal{A}$. But here we have $H(f) = G(f)$ and the result follows from the fact that G is exact. □

Lemma 3.7. Let $\mathcal{A}$, $\mathcal{B}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ be an exact functor. Let $\mathcal{C} \subset \mathcal{A}$ be a Serre subcategory contained in the kernel of F. Then $\mathcal{C} = \mathrm{Ker}(F)$ if and only if the induced functor $\overline{F} : \mathcal{A}/\mathcal{C} \to \mathcal{B}$ (Lemma 3.6) is faithful.

Proof. We will use the results of Lemma 3.6 without further mention. The "only if" direction is true because the kernel of $\overline{F}$ is zero by construction. Namely, if $f : X \to Y$ is a morphism in $\mathcal{A}/\mathcal{C}$ such that $\overline{F}(f) = 0$, then $\overline{F}(\mathrm{Im}(f)) = \mathrm{Im}(\overline{F}(f)) = 0$, hence $\mathrm{Im}(f) = 0$ by the assumption on the kernel of F. Thus $f = 0$.

For the "if" direction, let X be an object of $\mathcal{A}$ such that $F(X) = 0$. Then $\overline{F}(\mathrm{id}_X) = \mathrm{id}_{\overline{F}(X)} = 0$, thus $\mathrm{id}_X = 0$ in $\mathcal{A}/\mathcal{C}$ by faithfulness of $\overline{F}$. Hence $X = 0$ in $\mathcal{A}/\mathcal{C}$, that is $X \in \mathrm{Ob}(\mathcal{C})$. □

References

[1] Jean-Pierre Serre. Groupes d'homotopie et classes de groupes abéliens. Ann. of Math. (2), 58:258–294, 1953.